

Gridless Sparse Array Design for Beampattern Optimization

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Abstract—In integrated sensing and communication systems, the beampattern of a sensor array affects angular resolution in sensing applications and interference suppression in communication systems. Under constrained hardware costs, sparse arrays become an attractive option for achieving a large aperture with few sensors. However, many existing sparse arrays exhibit undesirable grating lobes in their beampatterns. The grid-based sensor locations in sparse arrays can also yield a high-complexity combinatorial array design. This paper presents a framework for the design of gridless sparse arrays. The array geometry is optimized using an objective function based on a kernel-windowed power pattern. The Fourier relation enables a modified objective function that is differentiable with respect to the gridless sensor locations. This property admits the gradient-based optimized array with spacing constraints (GBOA-SC) with a polynomial-time solution. Furthermore, a grid-based sparse array, denoted as R-GBOA-SC, is obtained by rounding the GBOA-SC to an integer grid. Numerical examples demonstrate the power patterns of GBOA-SC and R-GBOA-SC. In particular, GBOA-SC and R-GBOA-SC mitigate the grating lobes, compared with several array geometries.

Index Terms—Beampattern optimization, gridless sparse arrays, normalized gradient descent, Gaussian kernel functions.

I. INTRODUCTION

Sensor arrays play a crucial role in radar systems [1], [2], wireless communication systems [3], and integrated sensing and communication systems [4]. In these applications, the uniform linear array (ULA) with inter-element spacing d is ubiquitous due to its simplicity [5, Section 2.3]. Denoting λ as the wavelength of the incoming source signals, the N -sensor *dense* ULA with $d = \lambda/2$ has several properties. First, its beampattern exhibits a half-power beamwidth (HPBW) of order $\mathcal{O}(N^{-1})$ and sidelobe levels below -13 dB, which is important for resolving source waveforms from different directions [5, Chapter 2]. Second, the dense ULA can resolve $\mathcal{O}(N)$ source directions. However, for applications with a large aperture L [6], dense ULAs require a large number of elements. This attribute increases hardware costs and intensifies non-ideal effects such as mutual coupling [7, Section 8.7].

In this regard, *sparse arrays* with a small N and a large L have been widely studied in the literature. Sparse ULAs with $d > \lambda/2$ meet the specification of N and L . However, their

beampatterns suffer from grating lobes [7, Chapter 6]. The nested arrays (NAs) [8] and coprime arrays (CAs) [9], [10] utilize a large L and a small N with closed-form expressions for the sensor locations. Additionally, NAs and CAs can resolve more uncorrelated source directions than sensors with coarray processing [8]–[11]. However, the beampatterns of NAs and CAs still exhibit similar structures to the grating lobes. Minimum redundancy arrays (MRAs) [5], [12] optimize array geometry on integer grids to improve utilization of N and L . However, finding MRAs needs a combinatorial search, leading to high computational complexity for a large L .

Sparse array designs aiming for beampatterns have been studied in the literature. The binned random array [13, Section 19.2.4] achieves a lower power distribution near the steering direction but relies on heuristics. Recent works have addressed challenging optimization problems either by reformulating them into tractable forms [14]–[16] or by adopting machine-learning approaches [17], [18]. However, these approaches require advanced optimization tools.

This paper proposes an optimization framework for gridless sparse arrays. First, the association between the power pattern and the difference-coarray weight function leads to a Fourier relation, facilitating the derivation of an objective function from a kernel-smoothed difference-coarray weight function. For Gaussian kernels, the objective function has a closed-form and differentiable expression with respect to the gridless sensor locations. We propose the gradient-based optimized array with spacing constraints (GBOA-SC), which can be solved in polynomial time. Furthermore, rounding the GBOA-SC to an integer grid yields an array named R-GBOA-SC. Numerical examples demonstrate improved power patterns compared to several existing arrays. The direction-of-arrival (DOA) estimation performance is left for future work.

The remainder of this paper is organized as follows. Section II reviews the beampattern, power pattern, and difference coarray. Section III analyzes the objective function and develops the GBOA-SC and R-GBOA-SC. Section IV presents numerical examples of the power patterns of several array configurations. Section V concludes the paper.

II. PRELIMINARIES

We consider a linear array with N sensors. The sensors are located at $s_1\lambda/2, s_2\lambda/2, \dots, s_N\lambda/2$, where λ is the wavelength. This array is represented by the set $\mathbb{S} \triangleq$

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$\{s_1, s_2, \dots, s_N\} \subset \mathbb{R}$ with $0 = s_1 < s_2 < \dots < s_N = L$. The array aperture is L . The array position vector (APV) is defined as

$$\mathbf{s} \triangleq [s_1 \ s_2 \ \dots \ s_N]^\top. \quad (1)$$

The beam pattern of the array $\mathbb{S}$ in the angular domain is defined as [5, Section 2.2], [7, Section 6.3], [19, Section 2.4]:

$$B(\theta) \triangleq \frac{1}{N} \sum_{m=1}^N \exp(j s_m \pi \sin \theta), \quad (2)$$

where $j \triangleq \sqrt{-1}$ and $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ denotes the angle of arrival. Let $\Theta \triangleq \pi \sin(\theta) \in [-\pi, \pi]$ be the normalized angle. The power pattern of $\mathbb{S}$ is defined as [5, Section 2.6.1]

$$P(\Theta) \triangleq |B(\theta)|^2 = \frac{1}{N^2} \sum_{m,n=1}^N \exp(-j(s_m - s_n)\Theta). \quad (3)$$

It can be shown that $P(\Theta)$ has the maximum $P(\Theta) = 1$ if $\Theta = 0$. Due to (3), the power pattern depends purely on the differences between the elements in $\mathbb{S}$. This structure is specified by the difference coarray of $\mathbb{S}$, defined as [8], [9]

$$\mathbb{D}_2 \triangleq \{s_m - s_n \mid (m, n) \in \llbracket 1, N \rrbracket^2\}, \quad (4)$$

where the set $\llbracket 1, N \rrbracket \triangleq \{1, 2, \dots, N\}$. The difference-coarray weight function of each element in $\mathbb{D}_2$ is defined as the cardinality of element pairs with a specific difference ℓ [8]:

$$w(\ell) = \text{card}(\{(m, n) \in \llbracket 1, N \rrbracket^2 \mid s_m - s_n = \ell\}). \quad (5)$$

Substituting (5) into (3) results in another expression of the power pattern [20]:

$$P(\Theta) = \frac{1}{N^2} \sum_{\ell \in \mathbb{D}_2} w(\ell) \exp(-j\ell\Theta). \quad (6)$$

Equation (6) can be interpreted as a Fourier transform pair. If ℓ is an integer, then the power pattern $P(\Theta)$ can be regarded as the discrete-time Fourier transform (DTFT) of $w(\ell)/N^2$.

III. GRIDLESS SPARSE ARRAY DESIGN

A. Design Objectives for Gridless Sparse Arrays

Gridless sparse arrays have the real-valued difference ℓ in (5). The association between $P(\Theta)$ and $w(\ell)/N^2$ in (6) will be analyzed for this case.

To begin with, the continuous difference-coarray weight function $w_c(x)$ for $x \in \mathbb{R}$ is defined as the superposition of shifted Dirac delta functions, analogous to aperture sampling [5, Section 2.7.2]:

$$w_c(x) \triangleq \sum_{\ell \in \mathbb{D}_2} w(\ell) \delta(x - \ell), \quad (7)$$

Based on (6) and (7), the power pattern can be expressed as

$$P(\Theta) = \int_{-\infty}^{\infty} \frac{w_c(x)}{N^2} \exp(-jx\Theta) dx = \mathfrak{F} \left\{ \frac{w_c(x)}{N^2} \right\}. \quad (8)$$

Here, $\mathfrak{F}$ denotes the continuous-time Fourier transform (CTFT) [21, Chapter 4] from $x \in \mathbb{R}$ to $\Theta \in \mathbb{R}$. Equation (8) reveals the Fourier relation between $w_c(x)$ and $P(\Theta)$ up to a factor.

Gridless sparse arrays can be designed by extending the least-squares error pattern synthesis [5, Section 3.5] to the power pattern. We consider $P(\Theta) = 1$ at $\Theta = 0$ and the desired power pattern is 0 for $\Theta \neq 0$. The least-squares error pattern synthesis results in the error metric $\int_{-\infty}^{\infty} |P(\Theta) - 0|^2 d\Theta = \int_{-\infty}^{\infty} |P(\Theta)|^2 d\Theta$. However, this error metric is unbounded since the continuous difference-coarray weight function $w_c(x)$ has infinite energy.

For this reason, the Dirac delta function in (7) is replaced with a *finite-energy kernel* $\psi(x)$ with unit DC value $\int_{-\infty}^{\infty} \psi(x) dx = 1$. This approach enables the kernel-smoothed continuous difference-coarray weight function $\hat{w}_{c,\psi}(x)$, defined as

$$\hat{w}_{c,\psi}(x) \triangleq w_c(x) * \psi(x) = \sum_{\ell \in \mathbb{D}_2} w(\ell) \psi(x - \ell), \quad (9)$$

where $*$ denotes the convolution operator. The Fourier relation in (8) motivates the *kernel-windowed power pattern* $\hat{P}_\psi(\Theta)$, defined as the CTFT of $\hat{w}_{c,\psi}(x)$:

$$\hat{P}_\psi(\Theta) \triangleq \mathfrak{F} \left\{ \frac{\hat{w}_{c,\psi}(x)}{N^2} \right\} = \int_{-\infty}^{\infty} \frac{\hat{w}_{c,\psi}(x)}{N^2} \exp(-jx\Theta) dx. \quad (10)$$

The relation in (10) yields properties of the kernel version of the weight function and power pattern, as presented next.

Property 1: Let $\hat{w}_{c,\psi}(x)$ and $\hat{P}_\psi(\Theta)$ be defined in (9) and (10), respectively. The CTFT of the kernel $\psi(x)$ is denoted by $\Psi(\Theta)$. Then the following properties hold:

- 1) $\hat{P}_\psi(\Theta) = P(\Theta)\Psi(\Theta)$.
- 2) The DC values are $\Psi(0) = 1$, $P(0) = 1$, and $\hat{P}_\psi(0) = 1$.
- 3) The energy of the kernel-windowed power pattern is finite and related to the kernel-smoothed difference-coarray weight function by

$$\int_{-\infty}^{\infty} |\hat{P}_\psi(\Theta)|^2 d\Theta = \frac{2\pi}{N^4} \int_{-\infty}^{\infty} |\hat{w}_{c,\psi}(x)|^2 dx. \quad (11)$$

The results in Property 1 follow from the properties of the CTFT [21]. Furthermore, Property 1-1 interprets $\hat{P}_\psi(\Theta)$ as smoothing the power pattern $P(\Theta)$ with the kernel $\Psi(\Theta)$. Property 1-2 ensures the unit response of the power pattern at $\Theta = 0$. Property 1-3 associates the energy of $\hat{w}_{c,\psi}(x)$ with that of $\hat{P}_\psi(\Theta)$. Both quantities are finite.

The integral $\int_{-\infty}^{\infty} |\hat{P}_\psi(\Theta)|^2 d\Theta$ in (11) is regarded as the *kernel-windowed error metric* derived from $\int_{-\infty}^{\infty} |P(\Theta)|^2 d\Theta$ in the least-squares error pattern synthesis. The finiteness of the integrals in (11) guarantees the well-definiteness of the kernel-windowed error metric. Additionally, (11) indicates that minimizing the kernel-windowed error metric for a fixed N is equivalent to minimizing the objective function

$$J_\psi(\mathbf{s}) \triangleq \int_{-\infty}^{\infty} |\hat{w}_{c,\psi}(x)|^2 dx, \quad (12)$$

where $\mathbf{s}$ denotes the APV in (1).

The objective function in (12) is simplified next for computational purposes. According to (5), the difference ℓ can be

expanded into $\Delta_{m,n} \triangleq s_m - s_n$ or $\Delta_{p,q} \triangleq s_p - s_q$ for the indices $1 \leq m, n, p, q \leq N$. Substituting (9) into (12) yields

$$J_\psi(\mathbf{s}) = \sum_{m,n,p,q=1}^N \int_{-\infty}^{\infty} \psi(x - \Delta_{m,n}) \psi(x - \Delta_{p,q}) dx. \quad (13)$$

Note that (13) applies to a finite-energy kernel function $\psi(x)$. However, simplifying the integral in (13) requires explicit expressions of the kernel function. For simplicity, the kernel function is selected to be a Gaussian function, defined as

$$\psi_G(x) \triangleq \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{x^2}{2\sigma^2}\right), \quad (14)$$

where the standard deviation $\sigma > 0$. Substituting (14) into (13) results in an expression of the objective function

$$J_G(\mathbf{s}) = \frac{1}{2\sqrt{\pi}\sigma} \sum_{m,n,p,q=1}^N \exp\left(-\frac{\Delta_{m,n,p,q}^2}{4\sigma^2}\right), \quad (15)$$

where the subscript ‘‘G’’ stands for the Gaussian kernel function. The fourth-order difference of $\mathbb{S}$ is

$$\Delta_{m,n,p,q} \triangleq \Delta_{m,n} - \Delta_{p,q} = s_m - s_n - s_p + s_q. \quad (16)$$

The objective function (15) is non-convex.

Some remarks on $\Delta_{m,n,p,q}$ are made next. The fourth-order difference $\Delta_{m,n,p,q}$ has been known in sparse array design for *cumulant-based DOA estimation* [22], [23]. This topic designs the fourth-order difference coarray $\mathbb{D}_4 \triangleq \{\Delta_{m,n,p,q} \mid (m,n,p,q) \in [1, N]^4\}$ with a long central consecutive segment. However, this approach assumes grid-based sparse arrays, making the design problem inherently combinatorial.

As a comparison, the objective function $J_G(\mathbf{s})$ in (15) is also related to the fourth-order difference $\Delta_{m,n,p,q}$. However, $J_G(\mathbf{s})$ is developed for the power pattern, leading to a different meaning in *beamforming*. Furthermore, the gridless assumption of $\mathbf{s}$ also makes $J_G(\mathbf{s})$ differentiable.

B. Gradient-Based Optimization with Spacing Constraints

Based on (15), this paper proposes an optimization problem for gridless sparse arrays of N sensors and aperture L :

$$\min_{\mathbf{s} \in [0, L]^N} J_G(\mathbf{s}) \quad (17a)$$

$$\text{subject to } s_m - s_{m-1} \geq d_{\min}, \quad \forall m = 2, 3, \dots, N, \quad (17b)$$

$$s_1 = 0, \quad s_N = L. \quad (17c)$$

The APV is denoted by $\mathbf{s}$, whose entries $s_1, s_2, \dots, s_N$ are confined to the continuous interval $[0, L]$. The spacing constraints (17b) enforce the increasing order of $s_1, s_2, \dots, s_N$. The parameter $d_{\min} > 0$ limits the minimum inter-element spacing in the array, thereby controlling the mutual coupling effects [7, Section 8.7]. The constraint (17c) preserves a constant aperture L . The quantities N , $d_{\min}$, and L are assumed to be given in (17).

The optimization problem (17) enables a gradient-based method [17], [18] with polynomial-time computational complexity. The normalized gradient descent (NGD) [24] is selected for stable update terms. Let $\mathbf{s}(k)$ denote the APV in the

k th iteration. The descent direction $\mathbf{d}(k)$ and the normalized descent direction $\bar{\mathbf{d}}(k)$ are defined as

$$\mathbf{d}(k) \triangleq -\frac{\partial J_G(\mathbf{s})}{\partial \mathbf{s}} \bigg|_{\mathbf{s}=\mathbf{s}(k-1)}, \quad \bar{\mathbf{d}}(k) \triangleq \frac{\mathbf{d}(k)}{\|\mathbf{d}(k)\|_2}, \quad (18)$$

where $\|\mathbf{d}(k)\|_2$ represents the ℓ_2 -norm of $\mathbf{d}(k)$. This norm normalizes the descent direction in $\bar{\mathbf{d}}(k)$. For $i = 2, 3, \dots, N-1$, the i th entry of the *candidate APV* $\tilde{\mathbf{s}}(k)$ can be updated via

$$[\tilde{\mathbf{s}}(k)]_i \triangleq [\mathbf{s}(k-1)]_i + \eta [\bar{\mathbf{d}}(k)]_i, \quad (19)$$

where η is the step size with a recommended value of unity.

The candidate APV $\tilde{\mathbf{s}}(k)$ is next matched with the constraints (17b) and (17c). Equation (17c) can be satisfied via

$$[\tilde{\mathbf{s}}(k)]_1 = 0, \quad [\tilde{\mathbf{s}}(k)]_N = L. \quad (20)$$

The constraint (17b) can be expressed as $s_{i-1} + d_{\min} \leq s_i \leq s_{i+1} - d_{\min}$. Based on the APVs $\mathbf{s}(k-1)$ and $\mathbf{s}(k)$, this condition becomes

$$s_{i,\min} \leq [\tilde{\mathbf{s}}(k)]_i \leq s_{i,\max}, \quad (21)$$

where $s_{i,\min} \triangleq [\mathbf{s}(k)]_{i-1} + d_{\min}$ and $s_{i,\max} \triangleq [\mathbf{s}(k-1)]_{i+1} - d_{\min}$. If (21) is satisfied, then $[\tilde{\mathbf{s}}(k)]_i$ is a valid value of the i th entry of $\mathbf{s}(k)$. For the clarity of presentation, Algorithm 1 summarizes the steps of the gradient-based optimization with spacing constraints (GBO-SC).

The derivative of (15) with respect to each sensor location involves only terms where the index appears in (m, n, p, q) , reducing the quadruple summation to a constant number of triple summations and yielding a complexity of $\mathcal{O}(N^3)$. Calculating the gradient over N sensors yields a computational complexity of $\mathcal{O}(N^4)$ for evaluating Line 2 in Algorithm 1. This complexity is lower than the $\mathcal{O}(L^N)$ complexity associated with a combinatorial search of grid-based arrays.

The proposed GBOA-SC also paves the way for grid-based sparse array design. From the GBOA-SC, a grid-based sparse array is obtained by *rounding the sensor positions of $\mathbf{s}_{\text{GBOA-SC}}$ to the nearest integers*. The i th entry of the APV for the rounded GBOA-SC (R-GBOA-SC) is defined as

$$[\mathbf{s}_{\text{R-GBOA-SC}}]_i \triangleq \text{round}([\mathbf{s}_{\text{GBOA-SC}}]_i), \quad (22)$$

where $\text{round}(\cdot)$ denotes the rounding function.

IV. NUMERICAL EXAMPLES

We compare the dense ULA [5], the sparse ULA [5], the CA [9], [10], the NA [8], the GBOA-SC, and the R-GBOA-SC. The number of sensors in these arrays is $N = 30$. The GBO-SC is initialized with the sparse ULA of $N = 30$ and $L = 203$. The input parameters in the GBO-SC are $\sigma = 1$, $\eta = 1$, $\varepsilon = 10^{-5}$, $K_{\max} = 500$, and $d_{\min} = 1$. Among these arrays, the GBOA-SC is gridless, while the remaining arrays are grid-based. Table I lists the profiles for the grid-based arrays.

Fig. 1 depicts the APV in Algorithm 1, where each trajectory represents the evolution of $[\mathbf{s}(k)]_i$ with respect to the iteration index k . The trajectories of $[\mathbf{s}(k)]_1$ and $[\mathbf{s}(k)]_N$ remain constant during the iteration, as in (17c). Furthermore,

Algorithm 1 Gradient-based optimization with spacing constraints (GBO-SC)

Require: $s(0)$, σ , η , $K_{\max}$, ε , and $d_{\min}$.

Ensure: The output APV $s_{\text{GBOA-SC}}$

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1: for  $k = 1$  to  $K_{\max}$  do
2:   Evaluate  $\mathbf{d}(k)$  based on (18).
3:   if  $\|\mathbf{d}(k)\|_2 < \varepsilon$  then
4:     Return  $s_{\text{GBOA-SC}} \triangleq s(k-1)$ 
5:   end if
6:   Evaluate  $\bar{\mathbf{d}}(k)$  according to (18).
7:   Compute the candidate APV  $\tilde{s}(k)$  via (19) and (20).
8:    $[s(k)]_1 \triangleq [\tilde{s}(k)]_1$  and  $[s(k)]_N \triangleq [\tilde{s}(k)]_N$ 
9:   for  $i = 2$  to  $N-1$  do
10:    if (21) holds then
11:       $[s(k)]_i \triangleq [\tilde{s}(k)]_i$ 
12:    end if
13:  end for
14: end for
15: Return  $s_{\text{GBOA-SC}} \triangleq s(K_{\max})$ 

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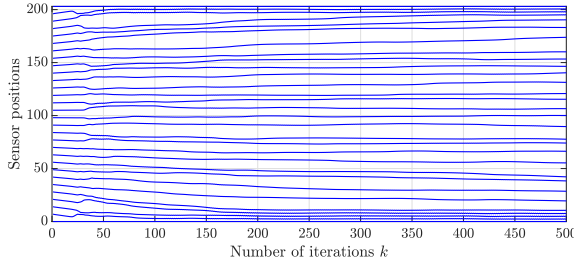


Fig. 1. Evolution of array geometry in computing the GBOA-SC.

all these trajectories are separated from each other, which satisfies (17b).

Fig. 2 illustrates the power patterns of the proposed GBOA-SC and the grid-based arrays for $N = 30$ under comparable L , as listed in Table I. The values of HPBW are also provided for a quantitative comparison. It is observed from Fig. 2(a) that the dense ULA has a wide HPBW and the decaying sidelobes with no grating lobes. The sparse ULA in Fig. 2(b) achieves a narrow HPBW at $\Theta = 0$ but suffers from grating lobes at $\Theta = \frac{2\pi}{7}, \frac{4\pi}{7}, \frac{6\pi}{7}$. These grating lobes result from the inter-element spacing $d > \lambda/2$ in the sparse ULA. The CA in Fig. 2(c) and the NA in Fig. 2(d) also exhibit narrow

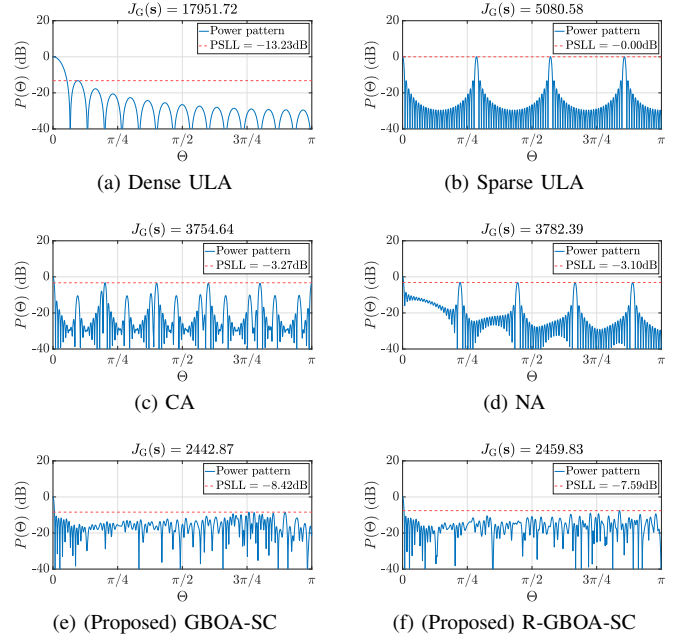


Fig. 2. The power patterns of the existing arrays and the proposed arrays.

HPBWs and grating lobes. These grating lobes arise from the segments $\{0, 10, 20, \dots, 200\}$, $\{21, 42, \dots, 189\}$ in the CA and $\{9, 18, 27, \dots, 198\}$ in the NA.

In contrast, the proposed GBOA-SC in Fig. 2(e) exhibits a narrow HPBW and a reduced peak sidelobe level compared with the sparse ULA, CA, and NA. This enhancement originates from the objective function $J_G(s)$ with the gridless s . Finally, the proposed R-GBOA-SC in Fig. 2(f) preserves the narrow HPBW and low sidelobes in the GBOA-SC. The objective function of the R-GBOA-SC is smaller than that of the sparse ULA, CA, and NA, but greater than that of the GBOA-SC. This degradation results from rounding the gridless GBOA-SC to the grid-based R-GBOA-SC in (22), where each element position may shift by up to 0.5 units.

V. CONCLUDING REMARKS

This paper proposed a new gridless array (GBOA-SC) and a new grid-based array (R-GBOA-SC) for beam pattern optimization. The GBOA-SC features a differentiable objective function derived from kernel-smoothed difference-coarray weight functions in the gridless setting. This formulation admits gradient-based algorithms for the GBOA-SC and rounding operations for the R-GBOA-SC. The GBOA-SC and R-GBOA-SC empirically exhibit narrow HPBWs and low peak sidelobe levels, compared with some existing arrays.

Future research aims to investigate the influence of initial arrays and parameters of kernel functions. The proposed objective function $J_\psi(s)$ may be extended to planar arrays, enabling further spatial sensing and communication applications.

VI. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

TABLE I
PROFILES FOR GRID-BASED ARRAY GEOMETRIES WITH $N = 30$

Array	Sensor locations	L	HPBW
Dense ULA	$\{0, 1, 2, \dots, 29\}$	29	0.1856
Sparse ULA	$\{0, 7, 14, 21, \dots, 203\}$	203	0.0265
CA	$\{0, 10, 20, \dots, 200\} \cup \{21, 42, \dots, 189\}$	200	0.0273
NA	$\{1, 2, 3, \dots, 8\} \cup \{9, 18, 27, \dots, 198\}$	197	0.0244
(Proposed) R-GBOA-SC	$\{0, 2, 5, 7, 11, 19, 29, 38, 42, 55, 66, 74, 78, 90, 100, 107, 115, 121, 131, 140, 146, 153, 160, 174, 183, 190, 195, 198, 201, 203\}$	203	0.0234

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